



devices to read-only mode by default, preventing any changes to data as we talked about in the previous section.

And there's a new entrant in the game, which we talk about at the end of the article!

INCIDENT RESPONSE:

When a digital forensic investigator isn't analysing firmware images and memory dumps, the incident response team is the one on the frontlines. Now, even though the two words themselves pretty much are self-explanatory, they don't reveal the full picture. An incident happened, and a team responded? Eh, not quite. Worry not as we unravel the mystery.

The aftermath of any security breach or cyberattack on an organization is devastating and it's the responsibility of the incident response team to approach the incident in an organised manner and try to minimise the damages. Any IR team has a developed and standardised response plan, which is repeatable with full efficiency every time:

- **Preparation:** A proactive approach, to assess the weak points in your security framework and weed out the possible attack vectors and footholds.
- **Implementation:** Confirming the availability of proper IR tools that allow overwatch for assets and activities, with abilities to filter out false positives.
- **Assessment and triage:** Determining what was the incident, its impact and how it occurred.
- **Containment:** Attempts to minimise the damages that might be caused by the event and containing it to stop further spread. Air-gapping/isolating the affected part of the network to prevent further damage.
- **Recovery:** Going back to the state before the incident. Restoring system backups, removing affected files with cleaner ones and applying missing patches, to curb the weakness.

- **Post-incident activities:** Updating and hardening your system, reporting of the incident for reviewing in future and to have a documented approach for similar attacks in future.

Now, it is time to check out the tools that make all this possible for an incident responder.

1. CimSweep

A suite of CIM/WMI(Common Information Model/Windows Management Instrumentation) tools, CimSweep is a toolkit that allows incident response and efficient bug hunting operations. Mainly focussed on Windows, CIMSweep supports all the versions of the Microsoft OS and even allows remote functionalities. It is an agent-less tool, that means there's no need to deploy it on the server that requires it, hence reducing detection rate and feasibility costs. Working with powerful Powershell cmdlets, CimSweep allows gathering of details like registry key events, log entry events, services, processes etc. To further extend its abilities, CimSweep provides the facility to incident responders to write their own domain-specific functions which can perform better with known malicious files, bad registry keys, suspicious activity, etc.

2. GRR Rapid Response

This is Google's very own incident response framework, released around 9 years ago. Unlike CimSweep, GRR is agent-based and cross-platform. A highly scalable framework, GRR can be deployed to perform automated incident response, including memory analysis, client device monitoring, automated scheduling for recurring tasks, registry search and many more. GRR has a very strong malware analysis functionality, putting it ahead of its competitors.

3. TheHive:

An open-source and free security incident response platform, TheHive has quite a following in the industry.

Focusing on teamwork and collaboration, TheHive allows SoC (Security Operations Center) analysts and CERTs(Computer emergency response team) to do elaborate analysis and security orchestration, working as a team together and performing quality, timely investigation of the cases. A python API client is available that allows creating cases and forwarding them to TheHive as alerts from different sources. Alongside TheHive Project's supplementary tools are also available such as Cortex (bulk analysis automation tool) and Hippocampe (a threat feed aggregator), making TheHive a pretty powerful tool.

4. AlienVault:

A security information and event management(SIEM) tool, AlienVault provides threat responders and analysts a comprehensive and detailed view of the system they are monitoring. With a robust log and asset management, AV keeps track of every happening that is occurring in a system at any given time, and it can use information from other security tools to gain a better understanding and provide a stronger context to the logged information, making it easier to tackle the problems, if any. AlienVault was acquired by AT&T, giving rise to the Cyber-security department of one of the biggest telecom companies.

This wraps up our discussion of digital forensics and incident response. Threats are always looming over organisations and attackers are more powerful than ever. With new critical vulnerabilities being uncovered daily to the inherent human weaknesses, the task of defending corporations is a mammoth one in the industry and there aren't enough people who can manage the responsibility of such intimidating stakes. The only choice we have is to either stay one step ahead of the attackers, or let them win, with career-changing consequences and damages. So, what do you choose? 🚩